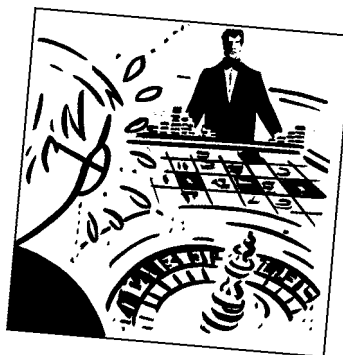


CHAPTER 29

Monte Carlo simulation



The aim of this Chapter...

...is to explain what is possibly the most useful technique for pricing derivatives and also for examining future scenarios in a probabilistic manner, the Monte Carlo simulation. You will see how the idea is applied to pricing hedged contracts and also to estimating the risks in unhedged positions.

In this Chapter...

- the relationship between option values and expectations for equities, currencies, commodities and indices
- the relationship between derivative products and expectations when interest rates are stochastic
- how to do Monte Carlo simulations to calculate derivative prices and to see the results of speculating with derivatives
- how to do simulations in many dimensions using Cholesky factorization

29.1 INTRODUCTION

The foundation of the theory of derivative pricing is the random walk of asset prices, interest rates, etc. We have seen this foundation in Chapter 4 for equities, currencies and commodities, and the resulting option pricing theory from Chapter 6 onwards. This is the Black–Scholes theory leading to the Black–Scholes parabolic partial differential equation. In Chapter 6 I mentioned the relationship between option prices and expectations. In this chapter we exploit this relationship, and see how derivative prices can be found from special simulations of the asset price, or interest rate, random walks. Briefly, the value of an option is the expected present value of the payoff. The catch in this is the precise definition of ‘expected.’

I am going to distinguish between the valuation of options having equities, indices, currencies, or commodities as their underlying with interest rates assumed to be deterministic and those products for which it is assumed that interest rates are stochastic. First, I show the relationship between derivative values and expectations with deterministic interest rates.

29.2 RELATIONSHIP BETWEEN DERIVATIVE VALUES AND SIMULATIONS: EQUITIES, INDICES, CURRENCIES, COMMODITIES

Recall from Chapter 6 that the fair value of an option in the Black–Scholes world is the present value of the expected payoff at expiry under a *risk-neutral* random walk for the underlying.

The risk-neutral random walk for S is

$$dS = rS dt + \sigma S dX.$$

We can therefore write

$$\text{option value} = e^{-r(T-t)} E [\text{payoff}(S)]$$

provided that the expectation is with respect to the risk-neutral random walk, not the *real* one.

This result leads to an estimate for the value of an option by following these simple steps:



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simulations

1. Simulate the risk-neutral random walk as discussed below, starting at today's value of the asset S_0 , over the required time horizon. This time period starts today and continues until the expiry of the option. This gives one realization of the underlying price path.
2. For this realization calculate the option payoff.
3. Perform many more such realizations over the time horizon.
4. Calculate the average payoff over all realizations.
5. Take the present value of this average, this is the option value.

29.3 GENERATING PATHS

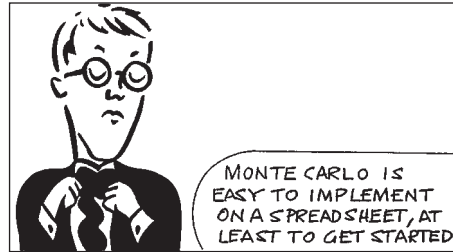
The initial part of this algorithm requires first of all the generation of random numbers from a standardized Normal distribution (or some suitable approximation). We discuss this issue below, but for the moment assume that we can generate such a series in sufficient quantities. Then one has to update the asset price at each time step using these random increments. Here we have a choice how to update S .

An obvious choice is to use

$$\delta S = rS \delta t + \sigma S \sqrt{\delta t} \phi,$$

where ϕ is drawn from a standardized Normal distribution. This discrete way of simulating the time series for S is called the **Euler method**. Simply put the latest value for S into the right-hand side to calculate δS and hence the next value for S . This method is easy to apply to any stochastic differential equation. This discretization method has an error of $O(\delta t)$.¹

The above algorithm is illustrated in the spreadsheet of Figure 29.1. The stock begins at time $t = 0$ with a value of 100, it has a volatility of 20%. The spreadsheet simultaneously



	A	B	C	D	E	F	G	H	I
1	Asset	100		Time	Sim1	Sim2	Sim3	Sim4	Sim5
2	Drift	5%		0	100.00	100.00	100.00	100.00	100.00
3	Volatility	20%		0.01	100.15	101.27	100.79	100.16	98.54
4	Timestep	0.01		0.02	99.80	100.84	102.72	102.31	101.66
5	Int.rate	5%		0.03	97.35	103.77	105.27	102.00	105.64
6				0.04	96.50	103.08	104.72	97.65	105.35
7			= D3+\$B\$4	0.05	101.25	101.61	102.37	102.76	103.63
8				0.06	97.53	100.49	104.47	106.86	99.04
9				0.07	97.11	102.00	107.70	105.72	99.20
10			=E3*EXP((B\$5-0.5*B\$3*B\$3)*B\$4+B\$3*SQRT(B\$4)*NORMSINV(RAND()))						8.77
11				0.09	85.74	100.79	109.07	106.01	97.95
12				0.1	81.32	100.99	105.13	105.40	100.32
94				0.92	102.25	105.44	88.51	96.74	96.08
95				0.93	100.68	105.48	90.44	97.04	95.36
96				0.94	102.26	104.01	92.40	99.26	94.67
97				0.95	102.10	103.47	88.99	95.27	97.09
98				0.96	100.11	103.36	88.95	92.74	96.30
99				0.97	101.34	104.06	89.26	93.59	97.19
100				0	= MAX(\$B\$104-F102,0)				25.29
101			=AVERAGE(E104:IV104)	0.99	103.71	102.73			58
102				1	104.94	104.47	91.86	95.05	98.79
103									
104	Strike	105	CALL	Payoff	0.00	0.00	0.00	0.00	0.00
105	= D105*EXP(-				Mean				
106	\$B\$5*\$D\$102)				PV				
107			PUT	Payoff	0.06	0.53	13.14	9.95	6.21
108			Mean		8.31				
109			PV		7.9				
110									
111									

Figure 29.1 Spreadsheet showing a Monte Carlo simulation to value a call and a put option.

¹ There are better approximations, for example the **Milstein method** which has an error of $O(\delta t^2)$.

calculates the values of a call and a put option. They both have an expiry of one year and a strike of 105. The interest rate is 5%. In this spreadsheet we see a small selection of a large number of Monte Carlo simulations of the random walk for S , using a drift rate of 5%. The time step was chosen to be 0.01. For each realization the final stock price is shown in row 102 (rows 13–93 have been hidden). The option payoffs are shown in rows 104 and 107. The mean of all these payoffs, over all the simulations, is shown in rows 105 and 108. In rows 106 and 109 we see the present values of the means, these are the option values. For serious option valuation you would not do such calculations on a spreadsheet. For the present example I took a relatively small number of sample paths.

The method is particularly suitable for path-dependent options. In the spreadsheet in Figure 29.2 I show how to value an Asian option. This contract pays an amount $\max(A - 105, 0)$ where A is the average of the asset price over the one-year life of the contract. The remaining details of the underlying are as in the previous example. How would the spreadsheet be modified if the average were only taken of the last six months of the contract's life?



Time Out...

Recap . . . risk neutrality

Remember, when using simulations for pricing *you must simulate the risk-neutral random walk*. The real random walk is theoretically irrelevant as far as option pricing is concerned. But once you've set up the code or spreadsheet for pricing one contract it can often be modified quite easily to price other, even path-dependent, contracts.

29.4 LOGNORMAL UNDERLYING, NO PATH DEPENDENCY

For the lognormal random walk we are lucky that we can find a simple, and exact, time stepping algorithm. We can write the risk-neutral stochastic differential equation for S in the form

$$d(\log S) = \left(r - \frac{1}{2}\sigma^2\right)dt + \sigma dX.$$

This can be integrated exactly to give

$$S(t) = S(0) \exp \left(\left(r - \frac{1}{2}\sigma^2\right)t + \sigma \int_0^t dX \right).$$

Or, over a time step δt ,

$$S(t + \delta t) = S(t) + \delta S = S(t) \exp \left(\left(r - \frac{1}{2}\sigma^2\right)\delta t + \sigma \sqrt{\delta t} \phi \right). \quad (29.1)$$

	A	B	C	D	E	F	G	H	I
1	Asset	100		Time	Sim1	Sim 2	Sim 3	Sim 4	Sim 5
2	Drift	5%		0	100.00	100.00	100.00	100.00	100.00
3	Volatility	20%		0.01	98.62	97.68	99.73	99.42	102.98
4	Timestep	0.01		0.02	100.69	96.45	101.13	101.28	101.36
5	Int.rate	5%		0.03	99.60	99.67	102.62	99.37	101.95
6				0.04	99.19	101.15	104.14	98.60	99.51
7		= D3+\$B\$4		0.05	104.10	100.00	105.03	98.97	96.86
8				0.06	104.71	99.11	103.22	96.93	98.89
9				0.07	103.07	95.00	101.00	96.00	99.03
10				0.08	107.03	100.00	100.00	99.00	99.00
11				0.09	110.57	100.93	101.59	99.34	98.75
12				0.1	114.50	100.24	99.36	95.67	99.88
13				0.11	114.43	101.32	100.22	94.92	102.90
93				0.91	101.02	111.09	119.38	77.82	85.23
94				0.92	101.54	109.58	118.06	80.13	83.75
95				0.93	101.38	108.20	118.49	79.96	83.54
96				0.94	103.38	107.87	119.79	82.21	83.12
97				0.95	107.58	108.43	116.24	81.58	84.69
98				0.97	107.20	112.81	115.56	83.07	90.72
99				0.98	109.18	113.45	116.22	82.50	87.23
100				0.99	110.49	114.04	116.22	82.50	87.23
101				1	113.23	117.67	120.05	81.49	93.27
102									
103									
104									
105									
106	Strike	105	ASIAN	Payoff	0.98	1.95	4.21	0.00	0.00
107			Mean	4.79					
108			PV	4.55					
109									
110									

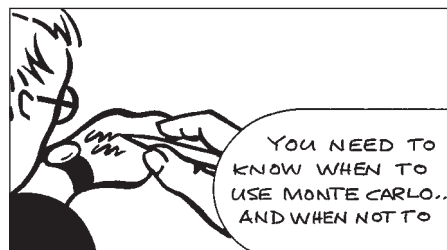
Figure 29.2 Spreadsheet showing a Monte Carlo simulation to value an Asian option.

Note that δt need not be small, since the expression is exact. Because this expression is exact and simple it is the best time stepping algorithm to use. Because it is exact, if we have a payoff that only depends on the final asset value, i.e. is European and path independent, then we can simulate the final asset price in one giant leap, using a time step of T .

If the option is path dependent then we have to go back to smaller time increments generally.

29.5 ADVANTAGES OF MONTE CARLO SIMULATION

Now that we have some idea of how Monte Carlo simulations are related to the pricing of options, I'll give you some of the benefits of using such simulations:



- The mathematics that you need to perform a Monte Carlo simulation can be very basic.
- Correlations can be easily modeled.

- There is plenty of software available, at the least there are spreadsheet functions that will suffice for most of the time.
- To get a better accuracy, just run more simulations.
- The effort in getting *some* answer is very low.
- The models can often be changed without much work.
- Complex path dependency can often be easily incorporated.
- People accept the technique and will believe your answers.

29.6 USING RANDOM NUMBERS

The Black–Scholes theory as we have seen it has been built on the assumption of either a simple up-or-down move in the asset price, the binomial model, or a Normally distributed return. When it comes to simulating a random walk for the asset price it doesn't matter very much what distribution we use for the random increments as long as the time step is small and thus that we have a large number of steps from the start to the finish of the asset price path. All we need are that the variance of the distribution must be finite and constant. (The constant must be such that the *annualized* volatility, i.e. the annualized standard deviation of returns, is the correct value. In particular, this means that it must scale with $\delta t^{1/2}$.) In the limit as the size of the time step goes to zero the simulations have the same probabilistic properties over a finite timescale regardless of the nature of the distribution over the infinitesimal timescale. This is a result of the central limit theorem.

Nevertheless, the most accurate model is the lognormal model with Normal returns. Since one has to worry about simulating sufficient paths to get an accurate option price one would ideally like not to have to worry about the size of the time step too much. As I said above, it is best to use the exact expression (29.1) and then the choice of time step does not affect the accuracy of the random walk. In some cases we can take a single time step since the time stepping algorithm is exact. If we do use such a large time step then we must generate Normally distributed random variables. I will discuss this below, where I describe the Box–Muller method.

If the size of the time step is δt then, for more complicated products, such as path-dependent ones, we may still introduce errors of $O(\delta t)$ by virtue of the discrete approximation to continuous events. An example would be of a continuous barrier. If we have a finite time step we miss the possibility of the barrier being triggered between steps. Generally speaking, the error due to the finiteness of the time step is $O(\delta t)$.

Because we are only simulating a finite number of an infinite number of possible paths, the error due to using N , say, realizations of the asset price paths is $O(N^{-1/2})$.

The total number of calculations required in the estimation of a derivative price is then $O(N/\delta t)$. This is then also a measure of the time taken in the calculation of the price. The error in the price is

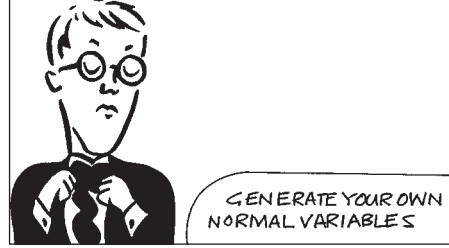
$$O\left(\max\left(\delta t, \frac{1}{\sqrt{N}}\right)\right),$$

i.e. the worst out of either the error due to the discreteness of the time step or the error in having only a finite number of realizations. To minimize this quantity, while keeping the total computing time fixed such that $O(N/\delta t) = K$, we must choose

$$N = O(K^{2/3}) \quad \text{and} \quad \delta t = O(K^{-1/3}).$$

29.7 GENERATING NORMAL VARIABLES

Some random number generators are good, others are bad, repeating themselves after a finite number of samples, or showing serial autocorrelation. Then they can be fast or slow. A particularly useful distribution that is easy to implement on a spreadsheet, and is fast, is the following approximation to the Normal distribution:



$$\left(\sum_{i=1}^{12} \psi_i \right) - 6,$$

where the ψ_i are independent random variables, drawn from a uniform distribution over zero to one. This distribution is close to Normal, having a mean of zero, a standard deviation of one, and a third moment of zero. It is in the fourth and higher moments that the distribution differs from Normal. I would use this in a spreadsheet when generating asset price paths with smallish time steps.

To get a better approximation you can just add up more uniform random variables. The general formula is

$$\sqrt{\frac{12}{N}} \left(\left(\sum_{i=1}^N \psi_i \right) - \frac{N}{2} \right).$$

This has a mean of zero and a standard deviation of one. As N gets larger so the approximation gets better. But if N gets too large we may as well use the following technique to produce genuinely Normal numbers.

29.7.1 Box–Muller

If you need to generate genuinely Normally distributed random numbers then the simplest technique is the **Box–Muller method**. This method takes uniformly distributed variables and turns them into Normal. The basic uniform numbers can be generated by any number of methods, see Press *et al.* (1992) for some algorithms. The Box–Muller method takes two uniform random numbers x_1 and x_2 between zero and one and combines them to give two numbers y_1 and y_2 that are both Normally distributed:

$$y_1 = \sqrt{-2 \ln x_1} \cos(2\pi x_2) \quad \text{and} \quad y_2 = \sqrt{-2 \ln x_1} \sin(2\pi x_2).$$

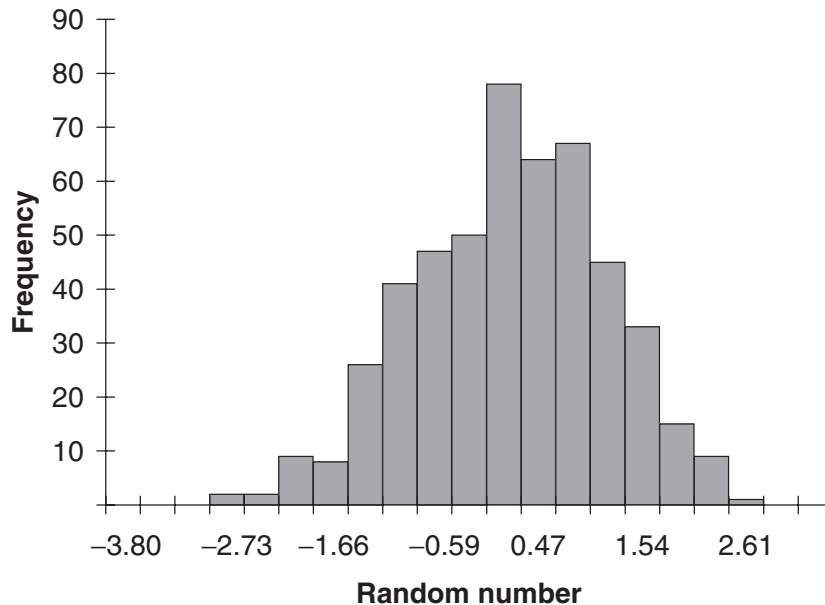


Figure 29.3 The approximation to the Normal distribution using 500 uniformly distributed points and the Box–Muller method.

Here is a Visual Basic function that outputs a Normally distributed variable.

```
Function BoxMuller()
Randomize
Do
    x = 2 * Rnd() - 1
    y = 2 * Rnd() - 1
    dist = x * x + y * y
Loop Until dist < 1
BoxMuller = x * Sqr(-2 * Log(dist) / dist)
End Function
```

In Figure 29.3 is the approximation to the Normal distribution using 500 points from the uniform distribution and the Box–Muller method.

For more efficient, more recent algorithms for generating Normal random variables see Jäckel (2002).

29.8 REAL VERSUS RISK NEUTRAL, SPECULATION VERSUS HEDGING

In Figure 29.4 are shown several realizations of a risk-neutral asset price random walk with 5% interest rate and 20% volatility. These are the thin lines. The bold lines in this figure are the corresponding *real* random walks using the same random numbers but here with a drift of 20% instead of the 5% interest rate drift. Although I am here emphasizing the use of Monte Carlo simulations in valuing options we can of course use them to estimate the payoff distribution from holding an *unhedged* option position. In this situation we are

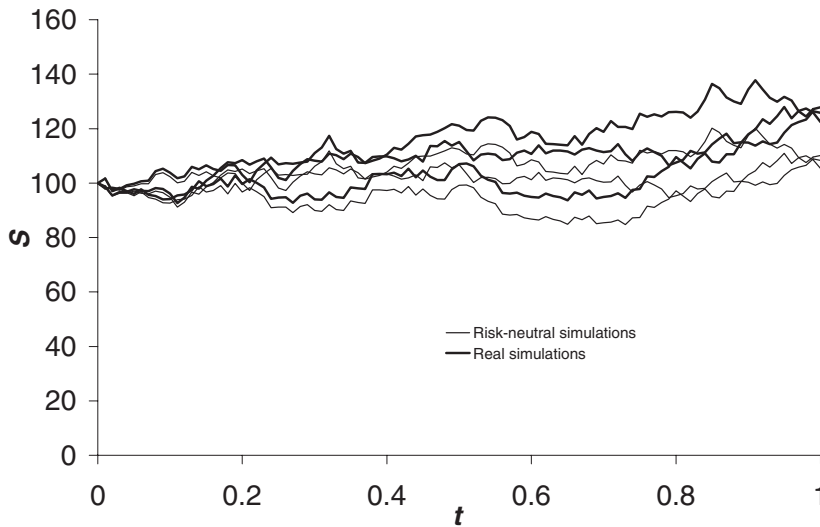


Figure 29.4 Several realizations of an asset price random walk.

interested in the whole distribution of payoffs (and their present values) and not just the average or expected value. This is because in holding an unhedged position we cannot guarantee the return that we (theoretically) get from a hedged position. It is therefore valid and important to have the real drift as one of the parameters; it would be incorrect to estimate the probability density function for the return from an unhedged position using the risk-neutral drift.

In Figure 29.5 I show the estimated probability density function and cumulative distribution function for a call with expiry one year and strike 105 using Monte Carlo simulations with $\mu = 20\%$ and $\sigma = 20\%$. The probability density function curve does not show the zero payoffs. The probability of expiring out of the money and receiving no payoff is approximately 25%.

In Figure 29.6 I show the estimated probability density function and cumulative distribution function for a put with the same expiry and strike, again using Monte Carlo simulations with $\mu = 20\%$ and $\sigma = 20\%$. The probability density function curve does not show the zero payoffs. The probability of expiring out of the money and receiving no payoff is approximately 75%.

Time Out...

Recap . . . reality

If you use simulations to get an idea of what may happen in the future to unhedged positions *use the real*



random walk for the underlying. You can estimate probability density functions, real expectations and standard deviations. These are all useful in risk management.

29.9 INTEREST RATE PRODUCTS

The relationship between expected payoffs and option values when the short-term interest rate is stochastic is slightly more complicated because there is the question of what rate to use for discounting.

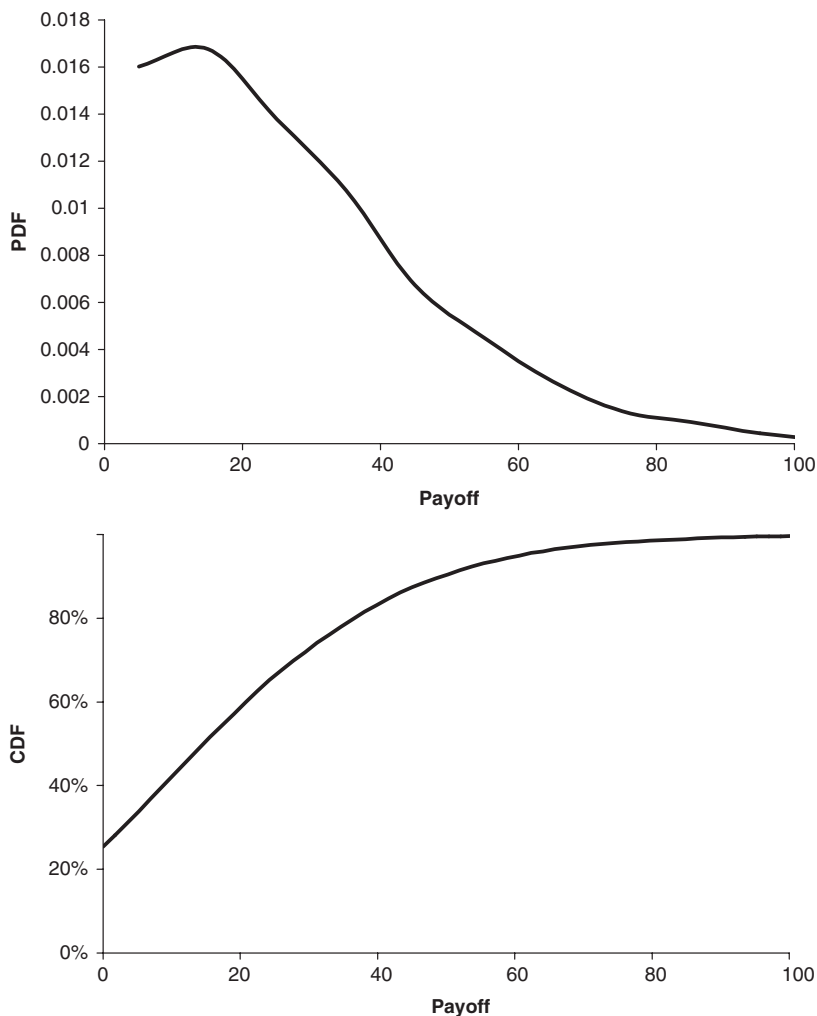


Figure 29.5 Real probability density function (top) and cumulative distribution function (bottom) for the payoff for a call.

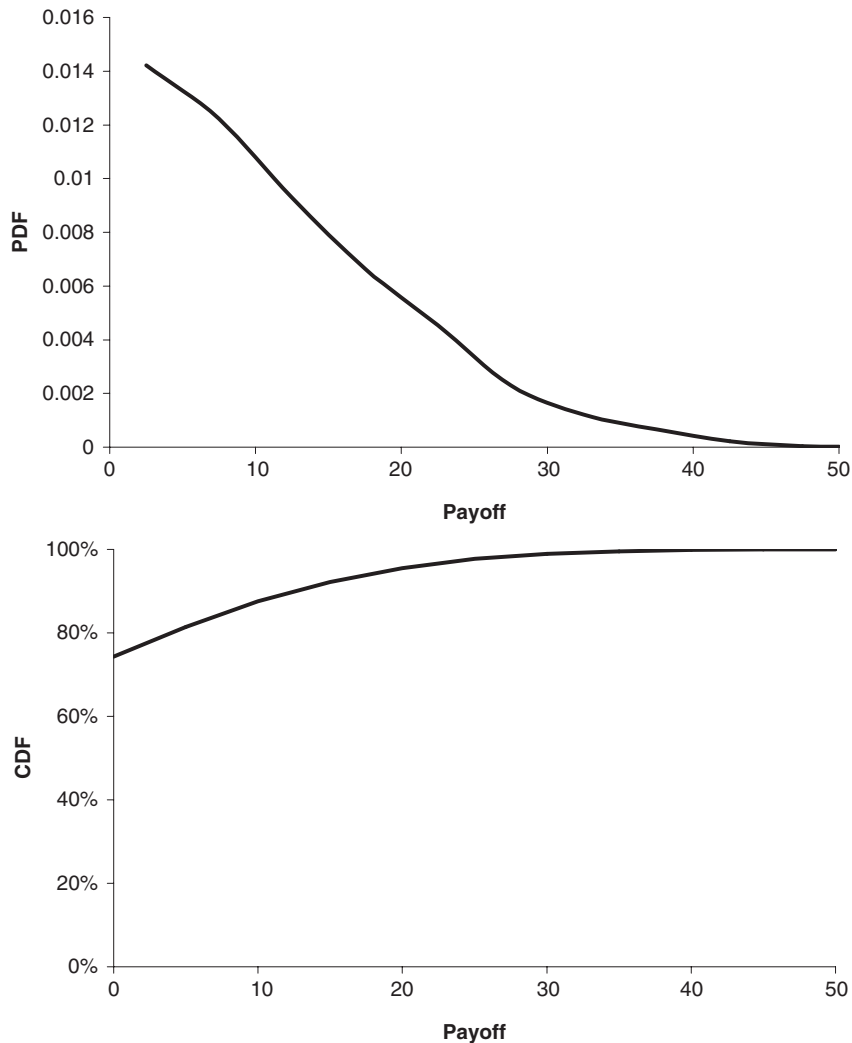


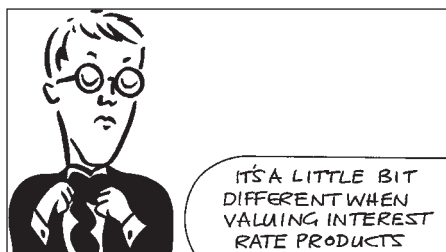
Figure 29.6 Real probability density function (top) and cumulative distribution function (bottom) for the payoff for a put.

The correct way to estimate option value with stochastic interest rates is as follows:

1. Simulate the random walk for the risk-adjusted spot interest rate r , as discussed below, starting at today's value of the spot rate, over the required time horizon. This time period starts today and continues until the expiry of the option. This gives one realization of the spot rate path.
2. For this realization calculate two quantities, the payoff and the *average* interest rate realized up until the payoff is received.



Pricing interest rate products



In other words,

$$\text{option value} = E \left[e^{-\int_t^T r(\tau) d\tau} \text{payoff}(r) \right].$$

Why is this different from the deterministic interest rate case? Why discount at the average interest rate? We discount all cashflows at the average rate because this is the interest rate received by a money market account, and in the risk-neutral world all assets have the same risk-free growth rate. Recall that cash in the bank grows according to

$$\frac{dM}{dt} = r(t)M.$$

The solution of which is

$$M(t) = M(T)e^{-\int_t^T r(\tau) d\tau}.$$

This contains the same discount factor as in the option value.

The choice of discretization of spot rate models is usually limited to the Euler method

$$\delta r = (u(r, t) - \lambda(r, t)w(r, t))dt + w(r, t)\sqrt{\delta t}\phi.$$

Rarely can the spot rate equations be exactly integrated.

In the next spreadsheet I demonstrate the Monte Carlo method for a contract with payoff $\max(r - 10\%, 0)$. Maturity is in one year. The model used to perform the simulations is Vasicek with constant parameters. The spot interest rate begins at 10%. The option value is the average present value in the last row.



Time Out...

Recap...

For pricing use the risk-neutral random walk; for analyzing an unhedged future use the real.

	AB		C	D	E	F	G	H
1	Spot rate	10%		Time	Sim 1	Sim2	Sim3	Sim4
2	Mean rate	8%		0	10.00%	10.00%	10.00%	10.00%
3	Reversion rate	0.2		0.01	9.99%	10.04%	10.10%	9.88%
4	Volatility	0.007		0.02	9.95%	10.07%	10.14%	9.87%
5	Timestep	0.01		0.03	9.88%	10.08%	10.08%	9.81%
6				0.04	9.78%	10.12%	10.18%	9.84%
7			= D3+\$B\$5	0.05	9.87%	10.17%	10.15%	9.75%
8				0.06	9.87%	10.18%	10.10%	9.82%
9				0.07	9.73%	10.20%	10.11%	9.81%
10				0.08	9.72%	10.21%	10.27%	9.82%
11				0.09	9.60%	10.22%	10.25%	9.73%
12	= F5+\$B\$3*(\$B\$2-						9%	9.79%
13	F5)*\$B\$5+\$B\$4*SQRT(\$B\$5)*(RAND()+RAND()+RAND()+RAND()+RAND()						4%	9.77%
14	+RAND()+RAND()+RAND()+RAND()+RAND()+RAND()+RAND()-6)						7%	9.75%
15				0.13	9.57%	10.15%	10.38%	9.73%
95				0.93	8.85%	10.44%	10.21%	9.27%
96	= AVERAGE(E2:E102)			0.94	8.81%	10.47%	10.23%	9.21%
97				0.95	8.83%	10.49%	10.17%	9.10%
98				0.96	8.81%	10.62%	10.19%	9.01%
99				0.97	8.73%	10.63%	10.29%	9.14%
100	= MAX(E102-\$B\$106,0)			0.98	8.68%	10.71%	10.28%	9.18%
101				0.99	8.56%	10.68%	10.15%	9.09%
102					8.42%	10.64%	10.33%	9.22%
103	= E106*EXP(-\$D\$102*E104)							
104					Mean rate	9.19%	10.32%	10.18%
105								
106	Strike	10%			Payoff	0.0000	0.0064	0.0033
107	= AVERAGE(E107:I107)					0.0000	0.0058	0.0030
108					PV'd			
109					Mean	0.001352		
110								
111								
112								
113								
114								

Figure 29.7 Spreadsheet showing a Monte Carlo simulation to value a contract with a payoff $\max(r - 10\%, 0)$.

29.10 CALCULATING THE GREEKS

The simplest way to calculate the delta of an option using Monte Carlo simulation is to estimate the option's value twice. The delta of the option is the derivative of the option with respect to the underlying

$$\Delta = \lim_{h \rightarrow 0} \frac{V(S + h, t) - V(S - h, t)}{2h}.$$

This is a central difference, discussed in Chapter 28. This is an accurate estimate of the first derivative, with an error of $O(h^2)$. However, the error in the measurement of the two option values at $S + h$ and $S - h$ can be much larger than this for the Monte Carlo simulation. These Monte Carlo errors are then magnified when divided by h , resulting in an error of $O(1/hN^{1/2})$. To overcome this problem, estimate the value of the option at $S + h$ and $S - h$ using the same values for the random numbers. In this way the errors in the Monte Carlo simulation will cancel each other out. The same principal is used to calculate the gamma and the theta of the option.

If we are dealing with a lognormal random walk then things are much simpler than this in practice. Simulate many lognormal random paths as usual, starting with stock at S . Now imagine that the stock had started at $(1 + \epsilon)S$. Do you have to resample the paths? No, because all you need do is multiple the final stock prices, at expiration, by the factor $1 + \epsilon$. Of course, this only works for a lognormal random walk. Now use these new, scaled, prices to calculate payoffs and hence an option value. Do the same but with a factor of $1 - \epsilon$. And then difference these two new option values and divide by $2\epsilon S$ and you have your delta, for almost no extra computational time.

Finally, we can also find the delta by observing what equation it satisfies.

The solution of the Black–Scholes partial differential equation

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0 \quad (29.2)$$

has the interpretation of being the expected value of the present value of the payoff, under a risk-neutral random walk.

The first ' r ' in this equation represents the ' r ' in

$$dS = rS dt + \sigma S dX.$$

The second ' r ' represents present valuing.

Differentiate Equation (29.2) with respect to S to get

$$\frac{\partial \Delta}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 \Delta}{\partial S^2} + (r + \sigma^2)S \frac{\partial \Delta}{\partial S} = 0,$$

where

$$\Delta = \frac{\partial V}{\partial S}.$$

A comparison with the Black–Scholes equation shows that the solution of this equation represents the expected value of the final Δ (a step function) under the random walk

$$dS = (r + \sigma^2)S dt + \sigma S dX.$$

We can use the same paths for S as used for calculating the option value but shifted according to

$$S e^{\int_0^t \sigma^2(\tau) d\tau}.$$

Now calculate the expected value of the step function (delta at expiration) using this shifted random walk. Since there is no discounting term in the partial differential equation there is no need to take the present value.

29.11 HIGHER DIMENSIONS: CHOLESKY FACTORIZATION

Monte Carlo simulation is a natural method for the pricing of European-style contracts that depend on many underlying assets. Supposing that we have an option paying off some function of $S_1, S_2, \dots, S_d$ then we could, in theory, write down a partial differential equation in $d + 1$ variables. Such a problem would be horrendously time consuming to

compute. The simulation methods discussed above can easily be extended to cover such a problem. All we need to do is to simulate

$$S_i(t + \delta t) = S_i(t) \exp \left(\left(r - \frac{1}{2} \sigma_i^2 \right) \delta t + \sigma_i \sqrt{\delta t} \phi_i \right).$$

The catch is that the ϕ_i are correlated,

$$E[\phi_i \phi_j] = \rho_{ij}.$$

How can we generate *correlated* random variables? This is where **Cholesky factorization** comes in.

Let us suppose that we can generate d *uncorrelated* Normally distributed variables $\epsilon_1, \epsilon_2, \dots, \epsilon_d$. We can use these variables to get correlated variables with the transformation

$$\phi = \mathbf{M}\epsilon \quad (29.3)$$

where ϕ and ϵ are the column vectors with ϕ_i and ϵ_i in the i th rows. The matrix $\mathbf{M}$ is special and must satisfy

$$\mathbf{M}\mathbf{M}^T = \Sigma$$

with Σ being the correlation matrix.

It is easy to show that this transformation will work. From (29.3) we have

$$\phi\phi^T = \mathbf{M}\epsilon\epsilon^T\mathbf{M}^T. \quad (29.4)$$

Taking expectations of each entry in this matrix equation gives

$$E[\phi\phi^T] = \mathbf{M}E[\epsilon\epsilon^T]\mathbf{M}^T = \mathbf{M}\mathbf{M}^T = \Sigma.$$

We can take expectations through the matrix multiplication in this because the right-hand side of (29.4) is linear in the terms $\epsilon_i\epsilon_j$.

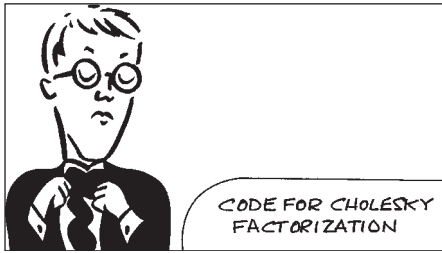
This decomposition of the correlation matrix into the product of two matrices is not unique. The Cholesky factorization gives one way of choosing this decomposition. It results in a matrix $\mathbf{M}$ that is lower triangular. Here is an algorithm for the factorization.

The matrix `Sigma` contains the correlation matrix with dimension `n`. The output matrix is contained in `M`.

```
Function cholesky(Sigma As Object)
Dim n As Integer
Dim k As Integer
Dim i As Integer
Dim j As Integer
Dim x As Double
Dim a() As Double
Dim M() As Double
n = Sigma.Columns.Count
ReDim a(1 To n, 1 To n)
ReDim M(1 To n, 1 To n)
```



Cholesky



```

For i = 1 To n
  For j = 1 To n
    a(i, j) = Sigma.Cells(i, j).Value
    M(i, j) = 0
  Next j
Next i
For i = 1 To n
  For j = i To n
    x = a(i, j)
    For k = 1 To (i - 1)
      x = x - M(i, k) * M(j, k)
    Next k
    If j = i Then
      M(i, i) = Sqr(x)
    Else
      M(j, i) = x / M(i, i)
    End If
  Next j
Next i
cholesky = M
End Function

```

One health warning, Cholesky decomposition can become unstable if you have any perfect correlation. To get more insight into this and for other decomposition methods see Jäckel (2002).

29.12 CALCULATION TIME

If we want an accuracy of ϵ and we have d underlyings (dimensions), how long will it take for us to get the price of an option?

Since the accuracy is proportional to the time step, δt , and inversely proportional to the square root of the number of paths, N , then we must have

$$\delta t = O(\epsilon) \quad \text{and} \quad N = O(\epsilon^{-2}).$$

The total calculation time is therefore

$$O(\delta t^{-1} \times N \times d) = O(\epsilon^{-1} \times \epsilon^{-2} \times d) = O(d\epsilon^{-3}).$$

The factor d appears because we will have to simulate d correlated assets.

Note that we don't get the greeks from this calculation. If we want to calculate the greeks then we will have to repeat the calculation as many more times as we need greeks. But, on the other hand, we can price many different options at the same time with little extra time cost, since the time is taken up generating the paths and not in calculating payoffs.

29.13 SPEEDING UP CONVERGENCE

Monte Carlo simulation is inefficient, compared with finite-difference methods, in dimensions fewer than about three. It is natural, therefore, to ask how can one speed up the convergence. There are several methods in common use, two of which I now describe.

29.13.1 Antithetic variables

In this technique one calculates two estimates for an option value using the one set of random numbers. We do this by using our Normal random numbers to generate one realization of the asset price path, an option payoff and its present value. Now take the same set of random numbers but change their signs, thus replace ϕ with $-\phi$. Again simulate a realization, and calculate the option payoff and its present value. Our estimate for the option value is the average of these two values. Perform this operation many times to get an accurate estimate for the option value.

This technique works because of the symmetry in the Normal distribution. This symmetry is guaranteed by the use of the antithetic variable.

29.13.2 Control variate technique

Suppose we have two similar derivatives, the former is the one we want to value by simulations and the second has a similar (but ‘nicer’) structure such that we have an explicit formula for its value. Use the *one* set of realizations to value *both* options. Call the values estimated by the Monte Carlo simulation V'_1 and V'_2 . If the accurate value of the second option is V_2 then a better estimate than V'_1 for the value of the first option is

$$V'_1 - V'_2 + V_2.$$

The argument behind this method is that the error in V'_1 will be the same as the error in V'_2 , and the latter is known.

A refinement of this technique is **martingale variance reduction**. In this method, one simulates one or more new dependent variables at the same time as the path of the underlying. This new stochastic variable is chosen so as to have an *expected value of zero* after each time step. This new variable, the ‘variate,’ is then added on to the value of the option. Since it has an expected value of zero it cannot make the estimate any worse, but if the variate is chosen carefully it can reduce the variance of the error significantly.

Let’s see how this is done in practice using a single variate. Suppose we simulate

$$\delta S = \mu S \delta t + \sigma S \sqrt{\delta t} \phi$$

to price our contract. Now introduce the variate y , satisfying

$$\delta y = f(S, t) (\delta S - E[\delta S]),$$

with zero initial value. Note that this has zero expectation. The choice of $f(S, t)$ will be discussed in a moment. The new estimate for the option value is simply

$$\bar{V} - \alpha e^{-r(T-t)} \bar{y},$$

where $\bar{V}$ is our usual Monte Carlo estimate and $\bar{y}$ is the average over all the realizations of the new variate at expiry. The choice of α is simple, choose it to minimize the variance of the error, i.e. to minimize

$$E \left[(V - \alpha e^{-r(T-t)} y)^2 \right].$$

I leave the details to the reader.

And the function $f(S, t)$? The natural choice is the delta of an option that is closely related to the option in question, one for which there is a closed-form solution. Such a choice corresponds to an approximate form of delta hedging, and thus reduces the fluctuation in the contract value along each path.

29.14 **PROS AND CONS OF MONTE CARLO SIMULATIONS**

The Monte Carlo technique is clearly very powerful and general. The concept readily carries over to exotic and path-dependent contracts, just simulate the random walk and the corresponding cash flows, estimate the average payoff and take its present value.

The main disadvantages are twofold. First, the method is slow when compared with the finite-difference solution of a partial differential equation. Generally speaking this is true for problems up to three or four dimensions. When there are four or more stochastic or path-dependent variables the Monte Carlo method becomes relatively more efficient. Second, the application to American options is far from straightforward. The reason for the problem with American options is to do with the optimality of early exercise. To know when it is optimal to exercise the option one must calculate the option price *for all values of S and t up to expiry* in order to check that at no time is there any arbitrage opportunity. However, the Monte Carlo method in its basic form is only used to estimate the option price at one point in S, t -space, now and at today's value.

Because Monte Carlo simulation is based on the generation of a finite number of realizations using series of random numbers, the value of a option derived in this way will vary each time the simulations are run. Roughly speaking, the error between the Monte Carlo estimate and the correct option price is of the order of the inverse square root of the number of simulations. More precisely, if the standard deviation in the option value using a single simulation is ϵ then the standard deviation of the error after N simulations is $\epsilon/\sqrt{N}$. To improve our accuracy by a factor of 10 we must perform 100 times as many simulations.

29.15 **AMERICAN OPTIONS**

Applying Monte Carlo methods to the valuation of European contracts is simple, but applying them to American options is harder. The problem is to do with the time direction in which we are solving. We have seen how it is natural in the partial differential equation framework to work backwards from expiry to the present. If we do this numerically then we find the value of a contract at every mesh point between now and expiry. This means that along the way we can ensure that there is no arbitrage, and in particular ensure that the early-exercise constraint is satisfied.

When we use the Monte Carlo method in its basic form for valuing a European option we only ever find the option's value at the one point, the current asset level and the current time. We have no information about the option value at any other asset level or time. So if our contract is American we have no way of knowing whether or not we violated the early-exercise constraint somewhere in the future.

In principle, we could find the option value at each point in asset-time space using Monte Carlo. For every asset value and time that we require knowledge of the option value we start a new simulation. But when we have early exercise we have to do this at

a large number of points in asset-time space, keeping track of whether the constraint is violated. If we find a value for the option that is below the payoff then we mark this point in asset-time space as one where we must exercise the option. And then for every other path that goes through this point we must exercise at this point, if not before. Such a procedure is possible, but the time taken grows exponentially with the number of points at which we value the option.

In Chapter 12 I said that there were no ‘good’ algorithms for Monte Carlo and early exercise. Well, that’s not strictly true anymore, as we’ll see in the next section.

29.16 LONGSTAFF & SCHWARTZ REGRESSION APPROACH FOR AMERICAN OPTIONS

There are several algorithms for pricing American options using Monte Carlo simulations, see Jäckel (2002) and Glasserman (2003) for details. But the one that is currently most popular with practitioners is that by Longstaff & Schwartz (2001). This method combines the forward simulation of stock price paths from startup to expiration, with the present valuing of cashflows along paths, but at each time step one looks at the benefit of exercising versus holding using a simple regression across stock prices. Let’s walk through the algorithm in detail with an example.

Consider an American put with strike price \$100, expiring in one year’s time. The stock is currently \$100, it has a volatility of 20% and the risk-free interest rate is 5%.

Step one, simulate many, N , realizations of the asset path from now to expiration using time steps of δt . If the random walk is lognormal then we will simulate

$$S_{j+1} = S_j \exp \left(\left(r - \frac{1}{2} \sigma^2 \right) \delta t + \sigma \sqrt{\delta t} \phi \right)$$

where ϕ is a Normally distributed random variable, and S_j is the stock price after j time steps. We start each path at $j = 0$ with $S_0 = 100$. In Figure 29.8 are shown 10 such paths, using five time steps from now to expiration. The numbers are shown in Figure 29.9.

Next step, calculate the payoffs at expiration for each of these 10 paths. These payoffs are shown in Figure 29.10. Notice that five paths end in the money and five out. Of course, this assumes that we haven’t exercised the put option at any time before expiration. We’ll get to that point in a moment. If we were to present value the average of all of these payoffs then we’d get the value of a European option.

The above tells us what we would receive if we exercised only at expiration, but we might have exercised earlier than that. Let’s go back to the fourth time step, time 0.8, and look at the decision to exercise at that time for each of the paths. In what follows I use the notation of Longstaff & Schwartz.

Figure 29.11 has an X and a Y column. The X column are the stock prices at time 0.8 *for those paths which are in the money at that time*. The Y column are the payoffs for these paths, discounted back from expiration to time 0.8. We want to calculate the cashflow from holding on to the option, conditional on the stock price at time 0.8. To do this we perform a regression of Y against a constant, X and X^2 . In this example² we find the least-squares fit to be

$$Y = -0.1472 X^2 + 25.347 X - 1075.2. \quad (29.5)$$

² Other regressions are possible and to be preferred. We’ll look at the subject of basis functions shortly.

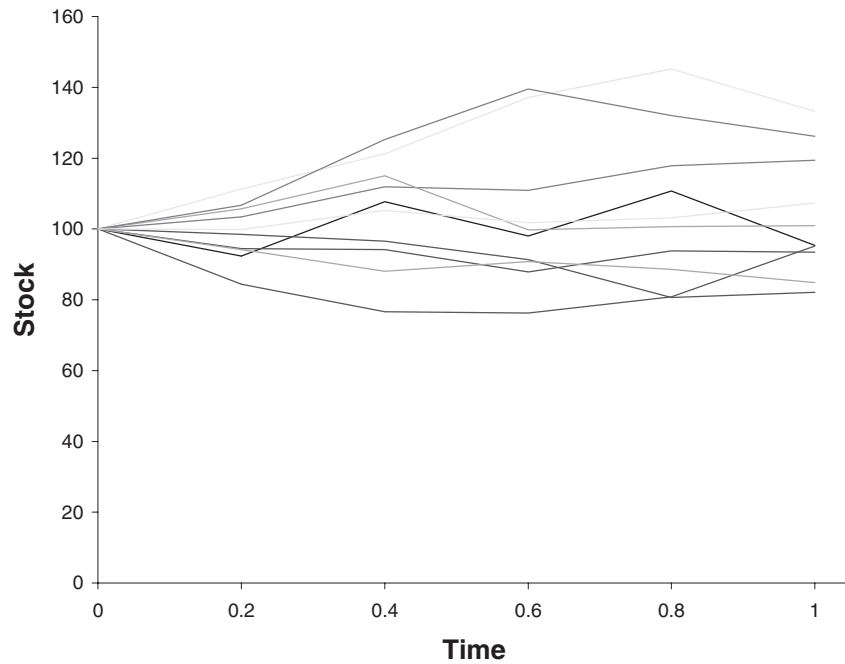


Figure 29.8 Ten stock price realizations.

Realization	0	0.2	0.4	0.6	0.8	1
1	100	92.30759	107.7357	98.04343	110.7416	95.34586
2	100	103.4446	111.9465	110.9322	117.8379	119.4419
3	100	111.2298	121.2417	137.1683	145.1687	133.1789
4	100	105.7152	115.0572	99.73054	100.6804	100.9471
5	100	98.47278	96.5825	91.32007	80.63689	82.1163
6	100	94.40168	94.16078	87.83702	93.84797	93.45847
7	100	106.7042	125.264	139.4822	132.0177	126.2041
8	100	84.37568	76.60055	76.21345	80.85454	95.19434
9	100	94.21698	88.00477	90.81541	88.63676	84.80556
10	100	99.81029	105.2631	101.747	103.1483	107.3703

Figure 29.9 The stock prices.

Realization	0	0.2	0.4	0.6	0.8	1	Payoff
1	100	92.30759	107.7357	98.04343	110.7416	95.34586	4.654138
2	100	103.4446	111.9465	110.9322	117.8379	119.4419	0
3	100	111.2298	121.2417	137.1683	145.1687	133.1789	0
4	100	105.7152	115.0572	99.73054	100.6804	100.9471	0
5	100	98.47278	96.5825	91.32007	80.63689	82.1163	17.8837
6	100	94.40168	94.16078	87.83702	93.84797	93.45847	6.541526
7	100	106.7042	125.264	139.4822	132.0177	126.2041	0
8	100	84.37568	76.60055	76.21345	80.85454	95.19434	4.805663
9	100	94.21698	88.00477	90.81541	88.63676	84.80556	15.19444
10	100	99.81029	105.2631	101.747	103.1483	107.3703	0

Figure 29.10 The stock prices and payoffs.

Realization	Y	X
1		
2		
3		
4		
5	0.99005 x 17.8837	80.63689
6	0.99005 x 6.541526	93.84797
7		
8	0.99005 x 4.805663	80.85454
9	0.99005 x 15.19444	88.63676
10		

Figure 29.11 Discounted payoffs and stock prices, for in the money at time 0.8 only. The discount factor is 0.99005.

Realization	Exercise now	Hold on
1		
2		
3		
4		
5	19.36311255	11.5635
6	6.152033497	7.109119
7		
8	19.14546028	11.90641
9	11.36323851	15.0028
10		

Figure 29.12 What you get from exercising versus holding on.

Now, for these same in-the-money-at-time-0.8 paths we compare the value of exercising immediately, time 0.8, with that from holding on using (29.5), see Figure 29.12. These numbers show that it is optimal to exercise now (time 0.8) for paths five and eight.

The next step is to build up a matrix of cashflows, as in Figure 29.13. This represents what is optimal *assuming that we don't exercise at any time step before 0.8*. Note that if there is a positive entry in the time 0.8 column it means that we should have exercised there (if not earlier) and so the cashflows for later times are set to zero.

Obviously we must continue to work backwards, next asking whether it would have been optimal to have exercised at time 0.6 for any of the paths. In Figure 29.14 are the cashflows, the Y column, present valued to time 0.6, and the stock prices, X, for those

Realization	0.2	0.4	0.6	0.8	1
1				0	4.654138
2				0	0
3				0	0
4				0	0
5				19.36311	0
6				0	6.541526
7				0	0
8				19.14546	0
9				0	15.19444
10				0	0

Figure 29.13 Cashflow matrix so far.

Realization	Y	X
1	0.980199 x 4.654138	98.04343
2		
3		
4	0	99.73054
5	0.99005 x 19.36311	91.32007
6	0.980199 x 6.541526	87.83702
7		
8	0.99005 x 19.14546	76.21345
9	0.980199 x 15.19444	90.81541
10		

Present value (at time 0.6) of the cashflows if we hold on.

Stock prices at time 0.6, but only for those paths which are **in the money at this time.**

Figure 29.14 Regression at time 0.6.

paths which are in the money at time 0.6. Note that there are now two discount factors because some cashflow has to be discounted one time step and others two time steps. The regression now gives

$$Y = -0.0361 X^2 + 5.6613 X - 203.95.$$

And off we go again, compare what you would get from exercising $\max(100 - X, 0)$ with the value from holding on $-0.0361 X^2 + 5.6613 X - 203.95$. After which we can draw up a new cashflow matrix.

Continue this process right back to time zero, and you should get the final cashflow matrix as shown in Figure 29.15. The last step in pricing the option is to present value all of these cashflows back to time zero and average. End result, 6.409. A finite-difference scheme gives the answer as 6.092.

Realization	0.2	0.4	0.6	0.8	1
1	0	0	0	0	4.6541375
2	0	0	0	0	0
3	0	0	0	0	0
4	0	0	0	0	0
5	0	0	0	19.363113	0
6	0	0	0	0	6.541526
7	0	0	0	0	0
8	0	0	23.786554	0	0
9	0	11.995234			0
10	0	0	0	0	0

Figure 29.15 Final cashflow matrix.

29.17 BASIS FUNCTIONS

I have used the simple quadratic basis function for the regression, since it is easiest to explain, and does a decent job. There are, however, better ones you can use. The original paper by Longstaff & Schwartz discusses this matter. When it comes to using this method in higher dimensions (which is the point after all) the choice of basis functions can become important.

29.18 SUMMARY

Simulations are at the very heart of finance. With simulations you can explore the unknown future, and act accordingly. Simulations can also be used to price options, although the future is uncertain, the result of hedging an option is theoretically guaranteed.

In this chapter I have tried to give a flavor of the potential of Monte Carlo and related methods. The reader should now be in a position to begin to use these methods in practice. The subject is a large and growing one, and the reader is referred to the section below for more information.

FURTHER READING

- See Boyle (1977) for the original application of Monte Carlo simulations to pricing derivatives.
- Duffie (1992) describes the important theory behind the validity of Monte Carlo simulations and also gives some clues about how to make the method efficient.
- The subject of Monte Carlo simulations is described straightforwardly and in detail by Vose (1997).
- For a review of Monte Carlo methods applied to American options see Boyle *et al.* (1995).
- For a very technical look at Monte Carlo see Jäckel (2002).

- For an American option algorithm see Longstaff & Schwartz (2001). This is a very good piece of work, nicely presented, understandable and with plenty of convincing examples. If only all papers were written like this.

EXERCISES

1. Simulate the risk-neutral random walk for an asset using a spreadsheet package, or otherwise. Use these data to calculate the value of a European call option.
2. Modify the spreadsheet in the above to output the option's delta.
3. Modify the above spreadsheet to value various exotic options such as barriers, Asians and lookbacks.
4. Why is it difficult to use Monte Carlo simulations to value American options?